SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

Name:Sri Janani B **Reg.No:192421036**

COURSE NAME: Computer Networks **COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Case study

QUESTIONS:4 Total Marks:40

1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss**, **180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

ANSWER:

1. Analyze whether the problem is primarily caused by the transport protocol or application layer.

The problem is mainly related to the **transport protocol and its interaction with the application layer**. Real-time video conferencing requires low delay, low jitter, and continuous data delivery.

The network shows **8% packet loss and 180 ms jitter**, which are very high for real-time communication. If TCP is used for real-time audio/video, lost packets are retransmitted. This can increase delay and cause buffering, frozen video, and delayed audio. TCP's reliability mechanism is therefore not ideal for time-sensitive multimedia traffic.

UDP is generally more suitable because it does not retransmit every lost packet and does not wait for missing packets before continuing transmission.

However, the application layer also plays an important role. Poor buffering, inappropriate jitter-buffer size, or lack of adaptive bitrate control can worsen the problem.

Conclusion: The primary issue is the unsuitable transport behavior combined with poor network conditions, while application-level buffering can further increase the impact.

2. Justify based on real-time multimedia requirements.

Real-time multimedia applications such as video conferencing require:

- **Low latency** – voice and video should arrive quickly.
- **Low jitter** – packets should arrive at relatively consistent intervals.
- **Low packet loss** – excessive loss causes missing audio/video.
- **Continuous transmission** – waiting for retransmissions can cause freezing.
- **Adaptive bandwidth usage** – video quality should adjust according to available bandwidth.

TCP provides reliable and ordered delivery, but retransmission and congestion-control mechanisms can introduce additional delay.

UDP does not provide guaranteed delivery, but this is acceptable for real-time multimedia because a slightly damaged or missing video frame is often better than waiting for an old packet to be retransmitted.

Therefore, protocols such as **UDP with RTP/RTCP** are commonly suitable for real-time media.

3. Recommended improvements

Protocol-level improvements:

1. Use **UDP with RTP/RTCP** for real-time audio/video.
2. Use congestion-control mechanisms suitable for real-time traffic.
3. Prioritize voice and video packets using **QoS**.
4. Use traffic classification and prioritization for conferencing traffic.
5. Monitor packet loss and jitter continuously.

Application-level improvements:

1. Use an adaptive jitter buffer.
2. Implement **adaptive bitrate streaming**.
3. Reduce video resolution when bandwidth decreases.
4. Use packet-loss concealment for audio/video.
5. Avoid excessive buffering because it increases latency.

Expected result: Lower jitter and delay, fewer freezes, and improved voice/video quality.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS**, **DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

ANSWER:

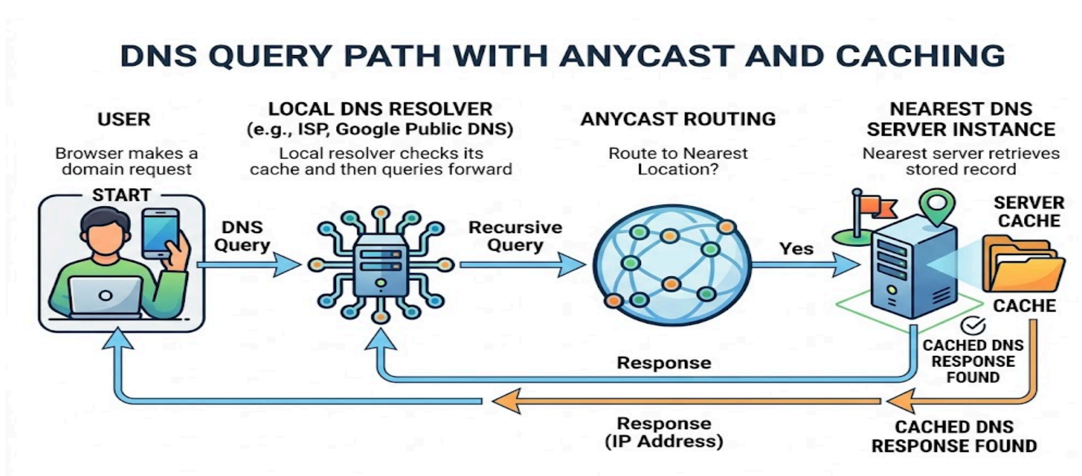
1. Analyze the reasons for high DNS resolution time.

The average DNS lookup time of **800 ms** is very high. Possible reasons include:

- **Geographical distance** – users may contact DNS servers located far away.
- **Single or centralized DNS servers** – excessive requests can overload the servers.
- **Multiple DNS lookups** – recursive resolution may require communication with several DNS servers.
- **Low TTL values** – cached DNS records expire quickly, causing more DNS queries.
- **DNS server overload** – heavy traffic increases response time.
- **Network congestion** – delays between clients and DNS servers increase lookup time.
- **Poor DNS caching strategy** – repeated queries may not be served from cache.
- Therefore, the organization needs geographically distributed DNS servers and efficient caching.

2. Proposed DNS architecture

A suitable architecture is:



The organization can deploy DNS servers at multiple global locations.

Anycast DNS

The same DNS IP address is advertised from multiple geographical locations using **BGP Anycast**.

For example:

- DNS Server – India
- DNS Server – Singapore
- DNS Server – Europe
- DNS Server – USA
- DNS Server – Australia

A user is automatically routed to a nearby/optimal DNS location.

★ **DNS pre-fetching**

Frequently accessed domain records can be pre-fetched before their TTL expires. This reduces cache misses and avoids unnecessary recursive DNS queries.

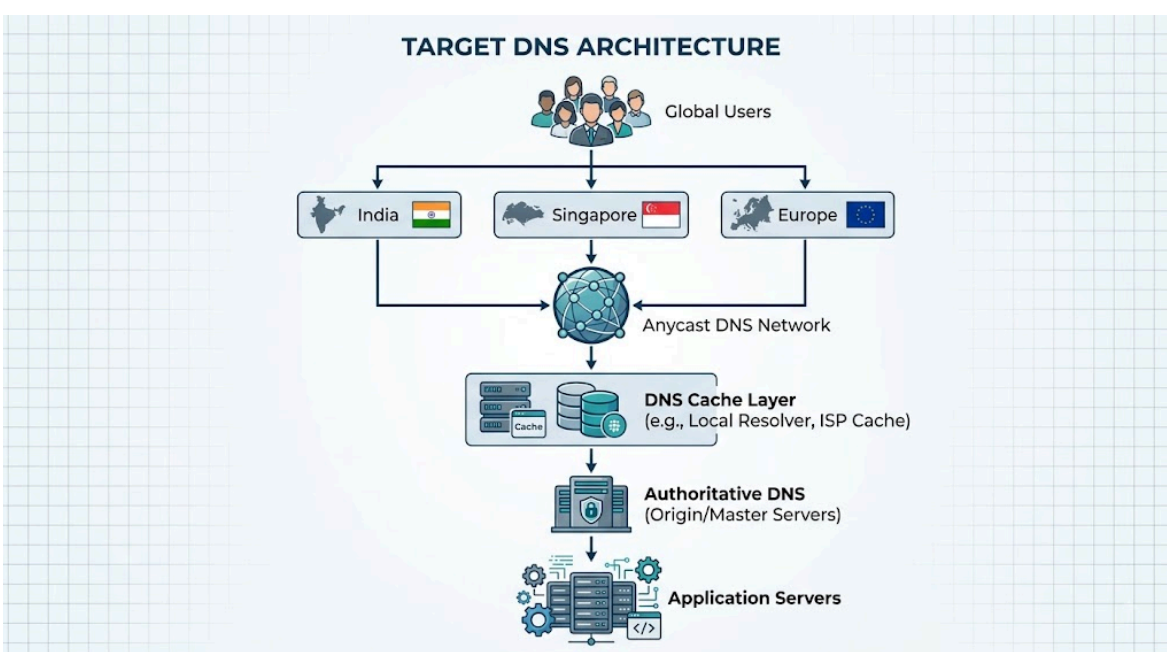
★ **TTL optimization**

A suitable TTL should be selected so that DNS records remain cached for a reasonable period.

- Very low TTL → frequent DNS queries.
- Very high TTL → slower propagation when IP addresses change.

A balanced TTL can significantly reduce DNS traffic.

Target architecture



With geographically distributed DNS, caching, Anycast, and optimized TTL values, the organization can target **sub-50 ms DNS response time for users located reasonably close to an edge resolver**. However, an absolute <50 ms cannot be guaranteed for every user and every lookup because network conditions vary.

3. Advantages and limitations

Factor	Advantages	Limitations
Anycast DNS	Low latency, automatic geographic routing	More complex infrastructure
DNS caching	Reduces repeated lookups	Stale records may remain until TTL expires
DNS pre-fetching	Reduces cache-miss delay	Generates additional background queries
TTL optimization	Balances performance and update speed	Incorrect TTL selection can cause problems
Distributed DNS	High availability and scalability	Higher operational cost

Conclusion: A distributed Anycast DNS architecture with caching, pre-fetching, and optimized TTL values provides better performance, scalability, and availability than a centralized DNS system.

3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

ANSWER:

1.Examine the impact of centralized DNS architecture.

The company currently sends all DNS requests to a **single primary DNS server**. This creates a major performance and availability problem.

When millions of users send DNS requests to one server:

- ➔ The server becomes a **bottleneck**.
- ➔ CPU and memory utilization increase.
- ➔ DNS response time increases.
- ➔ Users experience delays before applications start loading.
- ➔ A server failure can make DNS resolution unavailable.
- ➔ Promotional events create sudden traffic spikes.
- ➔ The approximately **800 ms DNS delay** directly increases application startup time.

For example:

User

|

| DNS Query

↓

Single DNS Server

|

| 800 ms response

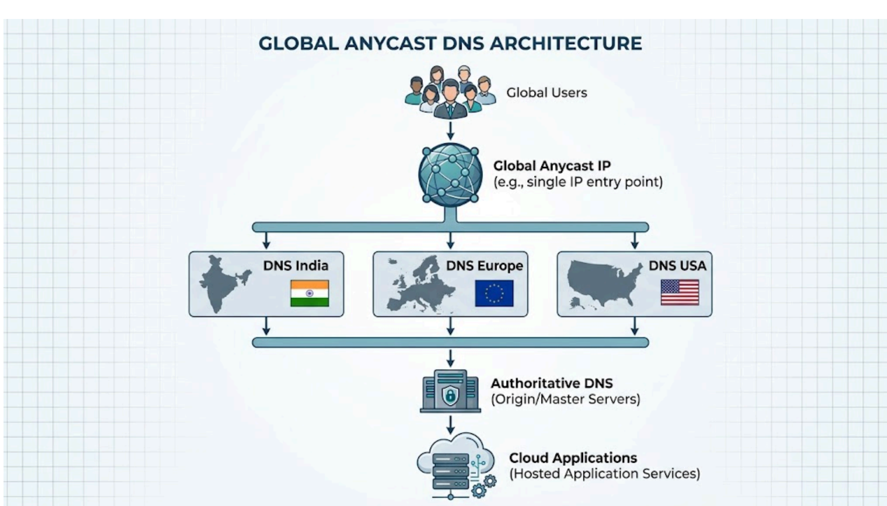
↓

Application Server

Even if the application server itself is fast, users must wait for DNS resolution before connecting to it.

2. Distributed DNS solution

A highly available solution can use **Anycast DNS with multiple geographically distributed DNS servers**.



Each DNS location advertises the same Anycast IP address.

When a user sends a DNS query, network routing directs the query toward an appropriate DNS location.

If one DNS location fails:

DNS India ❌

↓

Anycast Routing

↓

DNS Singapore / DNS Europe

↓

DNS Response

Thus, users can continue receiving DNS responses.

3. Impact of TTL, Anycast routing and DNS caching

TTL

TTL determines how long a DNS record can remain in a cache.

Low TTL:

- Faster DNS record changes.
- More DNS queries.
- Higher DNS infrastructure load.

High TTL:

- Fewer DNS queries.
- Better performance.
- Changes take longer to propagate.

Therefore, an appropriate TTL should be selected according to how frequently the application's IP addresses change.

Anycast routing

Anycast allows multiple DNS servers to use the same IP address.

Benefits:

- Routes users toward nearby DNS infrastructure.
- Reduces latency.
- Distributes traffic.
- Improves availability.
- Automatically provides alternate locations during failures.

DNS caching

Caching allows DNS resolvers to answer repeated queries without contacting authoritative DNS servers every time.

For example:

First request:

User → DNS → Authoritative Server

Later requests:

User → Cached DNS Response

This significantly reduces lookup time and DNS server workload.

Conclusion

A centralized DNS architecture is unsuitable for a global cloud service serving millions of users. A **distributed Anycast DNS architecture combined with DNS caching and carefully selected TTL values** provides lower latency, better scalability, and higher fault tolerance.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

ANSWER:

1. Analyze three TCP-related factors responsible for latency.

The order acknowledgment latency has increased from **1 ms to 40 ms**. The observed TCP retransmissions, longer connection establishment time, and excessive buffering indicate several TCP-related problems.

Factor 1 – TCP Flow Control

TCP uses a **receive window** to control how much data can be sent before receiving acknowledgments.

If the receiving application cannot process data quickly enough, the receive window may become small.

```
Sender →→→→ Receiver
        ↓
   Receive Window
        ↓
  Sender slows down
```

This can cause the sender to wait before transmitting additional data.

In a high-frequency trading system, even a small amount of waiting can significantly increase transaction latency.

Factor 2 – Nagle's Algorithm

Nagle's algorithm attempts to reduce the number of small TCP packets by combining small pieces of data before transmission.

This is useful for reducing network overhead, but it can be harmful to high-frequency transaction systems.

For example:

```
Order 1 → small packet
Order 2 → small packet
Order 3 → small packet
      ↓
Nagle waits to combine data
      ↓
Transmission delayed
```

For a trading application, the delay introduced by waiting to accumulate data may be more expensive than the additional network overhead.

Therefore, **TCP_NODELAY** may be considered when low latency is more important than reducing the number of small packets.

Factor 3 – SYN Queue Limitations

TCP establishes a connection using the **three-way handshake**:

```
Client → SYN
Server → SYN + ACK
Client → ACK
```

During market opening hours, thousands of connections may arrive simultaneously.

If the server's SYN backlog is too small:

- Incoming connection requests may be delayed.
- SYN packets may be retransmitted.
- Connection establishment takes longer.
- Application transactions are delayed.

This explains the observed increase in connection establishment time.

2. Effect on transaction processing

TCP Factor	Effect on Trading System
Flow control	Limits transmission when receiver buffers become full
Nagle's algorithm	Can delay small order messages
SYN queue limitation	Delays new TCP connections
TCP retransmissions	Increase acknowledgment and transaction latency
Excessive buffering	Creates additional waiting time

In high-frequency systems, these delays accumulate quickly. A transaction that normally takes **1 ms** may take **40 ms**, causing delayed order acknowledgments and potentially affecting trading decisions.

3. Recommended TCP configuration changes

1. Disable Nagle's algorithm where appropriate

Use:

TCP_NODELAY = enabled

This allows small order messages to be transmitted immediately instead of waiting to combine them.

2. Increase SYN backlog

Increase the server's SYN queue/backlog capacity so that large numbers of simultaneous connection requests can be handled during market opening.

3. Optimize TCP receive/send buffers

Configure TCP buffers according to the application's traffic pattern.

Avoid unnecessarily large buffers because excessive buffering can contribute to latency.

4. Use persistent TCP connections

Instead of creating a new TCP connection for every order:

Bad:

Order → TCP connection → close

Order → TCP connection → close

Prefer:

TCP Connection

↓

Order 1

Order 2

Order 3

Order 4

This avoids repeated three-way handshakes.

5. Monitor retransmissions

Investigate packet loss, congestion, interface errors, and network capacity because retransmissions directly increase latency.

6. Use appropriate congestion-control settings

The TCP congestion-control configuration should be tested and tuned for the organization's low-latency environment rather than blindly selecting a setting.

Final conclusion

The **40 ms latency** is likely caused by a combination of TCP flow-control limitations, Nagle-induced delays, SYN queue pressure, retransmissions, and excessive buffering. For a high-frequency trading platform, the recommended approach is to **use persistent connections, carefully tune TCP buffers, consider TCP_NODELAY, increase SYN backlog capacity, and eliminate the underlying causes of retransmissions**.

This combination can reduce unnecessary TCP delays while maintaining reliable order delivery.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand